

# Writing Small Numbers in Scientific Notation

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## MA.8.NSO.1.4

I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

# Warm-Up

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1. Write 4,300,000 in scientific notation.
2. Write  $10^{-3}$  as a fraction.
3.  $\sqrt{95}$  is between which two whole numbers?
4. Write 0.007 as a fraction.

# Warm-Up

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1. Write 4,300,000 in scientific notation.  $4.3 \times 10^6$
2. Write  $10^{-3}$  as a fraction.  $\frac{1}{1,000}$
3.  $\sqrt{95}$  is between which two whole numbers? **9 and 10**
4. Write 0.007 as a fraction.  $\frac{7}{1,000}$

# Notes

## I. Negative powers of ten name the decimal places

| $10^0$ | $10^{-1}$ | $10^{-2}$ | $10^{-3}$ | $10^{-4}$ | $10^{-5}$ |
|--------|-----------|-----------|-----------|-----------|-----------|
| 1      | 0.1       | 0.01      | 0.001     | 0.0001    | 0.00001   |

$$10^{-n} = \frac{1}{10^n} \qquad 10^{-3} = \frac{1}{1,000} = 0.001$$

A negative exponent puts the 1 in that decimal place:  $10^{-4}$  is a 1 in the fourth place.

# Notes

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## II. Small numbers in scientific notation

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$$0.000007 = 7 \times 10^{-6}$$

$$0.0000672 = 6.72 \times 10^{-5}$$

$$3.4 \times 10^{-9} = 0.0000000034$$

**The coefficient is still at least 1 and less than 10. The exponent is negative for a number less than 1.**

The exponent is how many places the point moves:  $0.0000672 \rightarrow 6.72$  is five places right, so  $10^{-5}$ .

# Notes

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## III. How many times larger — or smaller?

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$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}} = 4 \times 10^3 = 4,000$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 10^{12}$$

**Divide the coefficients. Divide the powers of 10 — subtract the exponents. Multiply the results.**

Which is bigger,  $10^{-5}$  or  $10^{-8}$ ? The one with the larger exponent:  $-5 > -8$ , so  $10^{-5}$ .

# Example 1

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The density of oxygen is 1,332 millionths of a gram per cubic centimeter.

Write this value as a decimal, then in scientific notation.

# Example 1

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$$\frac{1,332}{1,000,000} = 0.001332$$

$$0.001332 = 1.332 \times 10^{-3}$$

**Answer:  $0.001332 = 1.332 \times 10^{-3}$**

# Your Turn

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A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

# Your Turn

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A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

67 hundred-thousandths = 0.00067

Move the point 4 places right: 6.7

$$6.7 \times 10^{-4} = 0.00067$$

**Answer:  $6.7 \times 10^{-4}$**

## Example 2

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How many times larger is  $7.2 \times 10^{-4}$  than  $1.8 \times 10^{-7}$  ?

## Example 2

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$$\frac{7.2}{1.8} = 4$$

$$\frac{10^{-4}}{10^{-7}} = 10^{-4 - (-7)} = 10^3$$

$$4 \times 10^3 = 4,000$$

**Answer: 4,000 times larger**

# Your Turn

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How many times larger is  $5.2 \times 10^{-3}$  than  $2.6 \times 10^{-8}$  ?

# Your Turn

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How many times larger is  $5.2 \times 10^{-3}$  than  $2.6 \times 10^{-8}$  ?

Divide the numbers:  $5.2 \div 2.6 = 2$

Divide the powers:  $10^{-3} \div 10^{-8} = 10^{-3 - (-8)} = 10^5$

$2 \times 10^5 = 200,000$

**Answer: 200,000 times larger**

# Whiteboards · Question 1 of 9

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$$10^{-4}$$

**Answer it.**

# Whiteboards • Question 1 of 9

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$$10^{-4}$$
$$10^{-4} = \frac{1}{10^4} = \frac{1}{10,000} = 0.0001$$

**Answer: 0.0001**

# Whiteboards • Question 2 of 9

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Write 0.003 in scientific notation.

**Answer it.**

# Whiteboards • Question 2 of 9

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Write 0.003 in scientific notation.

The 3 is in the thousandths place:  $0.001 = 10^{-3}$

$$3 \times 10^{-3} = 0.003$$

**Answer:  $3 \times 10^{-3}$**

# Whiteboards • Question 3 of 9

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Write  $2.5 \times 10^{-4}$  in standard form.

**Answer it.**

# Whiteboards • Question 3 of 9

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Write  $2.5 \times 10^{-4}$  in standard form.

$$10^{-4} = 0.0001$$

$$2.5 \times 0.0001 = 0.00025$$

**Answer: 0.00025**

# Whiteboards • Question 4 of 9

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A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

**Write the tolerance in scientific notation.**

**Answer it.**

# Whiteboards · Question 4 of 9

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A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

**Write the tolerance in scientific notation.**

Move the point 5 places right: 9.88

$$9.88 \times 10^{-5} = 0.0000988$$

**Answer:  $9.88 \times 10^{-5}$**

# Whiteboards • Question 5 of 9

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Which is 0.00081 written in scientific notation?

**A.**  $8.1 \times 10^{-3}$

**B.**  $8.1 \times 10^4$

**C.**  $81 \times 10^{-5}$

**D.**  $8.1 \times 10^{-4}$

**Answer it.**

# Whiteboards • Question 5 of 9

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Which is 0.00081 written in scientific notation?

Move the point 4 places right: 8.1

$$8.1 \times 10^{-4} = 0.00081$$

**Answer: D —  $8.1 \times 10^{-4}$**

# Whiteboards • Question 6 of 9

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How many times larger is  $6 \times 10^{-3}$  than  $2 \times 10^{-5}$  ?

A. 300

B. 30

C. 3

D. 0.03

**Answer it.**

# Whiteboards • Question 6 of 9

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How many times larger is  $6 \times 10^{-3}$  than  $2 \times 10^{-5}$  ?

Divide the numbers:  $6 \div 2 = 3$

Divide the powers:  $10^{-3} \div 10^{-5} = 10^{-3 - (-5)} = 10^2$

$$3 \times 10^2 = 300$$

**Answer: A — 300**

# Whiteboards • Question 7 of 9

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Which is greater,  $3.9 \times 10^{-5}$  or  $7.4 \times 10^{-8}$  ?

**Answer it.**

# Whiteboards • Question 7 of 9

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Which is greater,  $3.9 \times 10^{-5}$  or  $7.4 \times 10^{-8}$  ?

$$3.9 \times 10^{-5} = 0.000039$$

$$7.4 \times 10^{-8} = 0.000000074$$

**Answer:  $3.9 \times 10^{-5}$**

# Whiteboards • Question 8 of 9

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A computer displays a number as  $4.2\text{E}-6$ .

Write the number in standard form.

**Answer it.**

# Whiteboards • Question 8 of 9

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A computer displays a number as  $4.2\text{E}-6$ .

Write the number in standard form.

$4.2\text{E}-6$  means  $4.2 \times 10^{-6}$

$4.2 \times 10^{-6} = 0.0000042$

**Answer: 0.0000042**

# Whiteboards • Question 9 of 9

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Fishing line A is  $2 \times 10^{-2}$  inch thick.

Fishing line B is  $6 \times 10^{-2}$  inch thick.

Which is thicker, and how many times? Name a thickness between them.

**Write your answer in sentences.**

# Whiteboards · Question 9 of 9

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Fishing line A is  $2 \times 10^{-2}$  inch thick.

Fishing line B is  $6 \times 10^{-2}$  inch thick.

Which is thicker, and how many times? Name a thickness between them.

Same power of 10, so compare 2 and 6.

$$6 \div 2 = 3$$

Between them:  $4 \times 10^{-2}$

**Answer: B; 3 times; e.g.  $4 \times 10^{-2}$  in.**

# Independent Set

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1. Write 0.00047 in scientific notation.
2. Write  $5.2 \times 10^{-3}$  in standard form.
3. Write 0.0000061 in scientific notation.
4. Which is greater,  $2.4 \times 10^{-6}$  or  $8.1 \times 10^{-9}$  ?
5. How many times larger is  $8 \times 10^{-2}$  than  $4 \times 10^{-5}$  ?
6. A computer displays a number as 3.7E-5. Write it in standard form.

# IXL

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1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Convert between standard and scientific notation: small numbers — 85M

Compare small numbers — KDR

Powers of ten with negative exponents — SKD